

Task 1: Formulating Your World as an MDP

1. MDP

The daily routine of a student deciding how to spend their time as a Markov Decision Process (MDP).

State Space (S)

$$S = \{\text{Fresh, Bored, Tired}\}$$

Action Space (A)

$$A = \{\text{Study, Play Games, Sleep}\}$$

Rewards (R)

State	Action	Reward
Fresh	Study	+5
Fresh	Play Games	-3
Fresh	Sleep	-5
Bored	Study	-3
Bored	Play Games	+5
Bored	Sleep	+3
Tired	Study	-5
Tired	Play Games	-3
Tired	Sleep	+5

Transition Probabilities (P)

Current State	Action	Next State Probabilities
Fresh	Study	Bored (0.7), Tired (0.3)
Fresh	Play Games	Tired (0.6), Bored (0.4)
Fresh	Sleep	Fresh (1.0)
Bored	Study	Tired (0.8), Bored (0.2)
Bored	Play Games	Fresh (0.8), Tired (0.2)
Bored	Sleep	Fresh (0.9), Bored (0.1)
Tired	Study	Tired (1.0)
Tired	Play Games	Tired (0.9), Bored (0.1)
Tired	Sleep	Fresh (0.8), Bored (0.2)

2. Value Iteration

- Discount factor: $\gamma = 0.9$
- Initialization: $V_0(s) = 0$ for all $s \in S$.

The Bellman optimality update rule is:

$$V_{k+1}(s) = \max(a) [R(s, a) + \gamma \sum P(s' | s, a) V_k(s')]$$

Iteration 1 (k = 1)

$$V_1(\text{Fresh}) = \max(5, -3, -5) = 5 \text{ (Action: Study)}$$

$$V_1(\text{Bored}) = \max(-3, 5, 3) = 5 \text{ (Action: Play Games)}$$

$$V_1(\text{Tired}) = \max(-5, -3, 5) = 5 \text{ (Action: Sleep)}$$

Iteration 2 (k = 2)

For s = Fresh:

$$a = \text{Study: } 5 + 0.9 \times [0.7(5) + 0.3(5)] = 5 + 0.9(5) = 9.5$$

$$a = \text{Play: } -3 + 0.9 \times [0.6(5) + 0.4(5)] = -3 + 0.9(5) = 1.5$$

$$a = \text{Sleep: } -5 + 0.9 \times [1.0(5)] = -5 + 0.9(5) = -0.5$$

$$V_2(\text{Fresh}) = \max(9.5, 1.5, -0.5) = 9.5 \text{ (Optimal Action - Study)}$$

For s = Bored:

$$a = \text{Study: } -3 + 0.9 \times [0.8(5) + 0.2(5)] = -3 + 0.9(5) = 1.5$$

$$a = \text{Play: } 5 + 0.9 \times [0.8(5) + 0.2(5)] = 5 + 0.9(5) = 9.5$$

$$a = \text{Sleep: } 3 + 0.9 \times [0.9(5) + 0.1(5)] = 3 + 0.9(5) = 7.5$$

$$V_2(\text{Bored}) = \max(1.5, 9.5, 7.5) = 9.5 \text{ (Optimal Action - Play Games)}$$

For s = Tired:

$$a = \text{Study: } -5 + 0.9 \times [1.0(5)] = -5 + 0.9(5) = -0.5$$

$$a = \text{Play: } -3 + 0.9 \times [0.9(5) + 0.1(5)] = -3 + 0.9(5) = 1.5$$

$$a = \text{Sleep: } 5 + 0.9 \times [0.8(5) + 0.2(5)] = 5 + 0.9(5) = 9.5$$

$$V_2(\text{Tired}) = \max(-0.5, 1.5, 9.5) = 9.5 \text{ (Optimal Action - Sleep)}$$